Kaylie Chen

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EDUCATION

University of British Columbia, Vancouver *September 2023 - Present*
Bachelor of Applied Science, Biomedical Engineering (Mechanical Specialization)
UBC Applied Science Co-op Program

KEY SKILLS

Mechanical Design: SolidWorks, Fusion 360, Onshape, FDM, GD&T, DFMA, CNC & Manual Milling
Software: GitHub, Git, Microsoft Office, Minitab, LTSpice, Arduino
Programming: Java, C, Python, R, MATLAB
Quality & Regulation: Medical Device V&V, Test Method Validation, Root Cause Analysis
Certifications: CASTL Intro to Biopharmaceutical Manufacturing, Standard First Aid CPR

TECHNICAL EXPERIENCE

Kardium Inc. — *Burnaby, BC*
Mechanical Test Engineering Intern *September 2026 - Present*

- Executed mechanical **verification and validation (V&V) testing** under approved SOPs, test protocols, and Test Method Validation (TMV) requirements to support product and design decisions
- Designed, machined, and assembled **custom test fixtures and jigs**, improving loading uniformity and sample alignment for test specific conditions
- Analyzed mechanical test data using statistical methods to assess result validity, variability, and significance, documenting findings compliant with **Good Documentation Practices (GDP)**

Sentact Technologies (UBC HATCH Ventures) — *Vancouver, BC*
Mechanical Engineering Intern *July 2026 - August 2026*

- Developed a **fully parametric workflow** using a Python dashboard, SolidWorks macros, and configurations for an AFP roller for carbon fiber layup, enabling rapid dimensional customization
- Designed custom 3D-printed moulds for polyurethane overmoulding, **reducing part defects by 25%**
- Validated functional roller prototypes with polyurethane casting and injection moulding, refining process parameters for **improved reproducibility and consistency**

TruMotion Technologies (UBC HATCH Ventures) — *Vancouver, BC*
Mechanical Engineering Intern *May 2026 - June 2026*

- Redesigned load-bearing components of a motor-driven rehabilitation device using FEA, **improving structural alignment and reducing mechanical backlash under operating loads of 900N**
- Defined and integrated **mechanical safety limits using human range-of-motion and biomechanical constraints**, preventing unsafe joint positions during rehabilitation
- Redesigned prototype components for manufacturable by developing **GD&T engineering drawings** and coordinating directly with manufacturers through fabrication

DESIGN TEAM EXPERIENCE

UBC Bionic Hand — *Vancouver, BC*
Mechanical Sub-Team Lead *September 2024 - Present*

- Led a 7-member sub-team through the 4th and 5th iterations of a prosthetic arm by defining system requirements, design objectives, and project ownership to **establish a mechanical design framework**
- Facilitated the development process through structured design reviews, milestone planning, and communication with electrical and controls teams, improving design accountability and integration
- Managed workload across parallel mechanical projects by prioritizing high-risk tasks and aligning ownership with member capacity, keeping critical designs within timeline
- Designed and prototyped a **gear-driven wrist rotation mechanism** using bevel gears and compliant linkages, enabling controlled actuation for a fully mechanical, body-powered prosthetic arm
- Performed **torque and load analysis** to size gears and validate component strength under applied wrist loads, supporting structural reliability and future motorized iterations